

Supplementary Material

Forecasting annual counts of large wildfires in Spain

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1 Reproducibility map

File	Purpose
<code>analisis_clasico_arima_arimax_v13.py</code>	Prespecified ARIMA grid, AutoReg models, exponential smoothing, Holt, ARIMAX lag sensitivity, fit and predictive R^2 , relative skill and sequential external validation.
<code>modelos_conteo_gif_v13.py</code>	NB2 observation-driven models, Bayesian negative-binomial AR(1), rolling-origin validation, probabilistic metrics and external validation.
<code>analysis_prospective_2026.py</code>	Frozen 2026 ARIMA prediction, dated MITECO snapshot, historical same-date completion benchmark and conditional lag-two heat-exposure scenarios.

2 Response overdispersion

For 2005–2023, $n = 19$, mean = 25.684, variance = 319.006, variance-to-mean ratio = 12.420, minimum = 3 and maximum = 58. The Bayesian parameterisation uses $\text{Var}(N \mid \mu, \alpha) = \mu + \mu^2/\alpha$, whereas the frequentist NB2 parameterisation uses $\mu + \phi\mu^2$.

3 Complete classical benchmark

The constant-only ARIMA(0, 0, 0) and the unstable highest-order specification were excluded a priori. Rolling-origin errors, rather than in-sample AICc alone, determined the benchmark.

3.1 ARIMA grid

Model	n	MAE	RMSE	NMAE	MASE	R^2_{pred}
ARIMA(1, 0, 0)	10	12.84	18.81	0.556	0.616	-0.177
ARIMA(2, 0, 0)	10	14.30	19.44	0.619	0.686	-0.258
ARIMA(0, 0, 1)	10	14.61	18.52	0.633	0.701	-0.142
ARIMA(2, 0, 1)	10	15.46	19.54	0.669	0.742	-0.271
ARIMA(0, 1, 1)	10	15.66	21.32	0.678	0.752	-0.512
ARIMA(2, 1, 1)	10	16.01	22.26	0.693	0.769	-0.649
ARIMA(1, 1, 1)	10	16.08	22.18	0.696	0.772	-0.637
ARIMA(1, 0, 1)	10	16.44	20.95	0.712	0.789	-0.461
ARIMA(0, 0, 2)	10	17.06	22.13	0.739	0.819	-0.630
ARIMA(3, 0, 0)	10	17.52	20.70	0.758	0.841	-0.426
ARIMA(1, 0, 2)	10	19.26	24.10	0.834	0.924	-0.932
ARIMA(2, 1, 0)	10	19.46	25.18	0.843	0.934	-1.111
ARIMA(2, 0, 3)	9	20.22	26.09	0.812	0.971	-1.255
ARIMA(2, 1, 2)	10	20.34	28.12	0.880	0.976	-1.632
ARIMA(3, 1, 2)	9	20.40	29.81	0.820	0.979	-1.944
ARIMA(0, 1, 2)	10	20.78	27.41	0.899	0.997	-1.500
ARIMA(1, 1, 0)	10	20.80	25.78	0.901	0.999	-1.211
ARIMA(3, 1, 0)	10	22.27	30.37	0.964	1.069	-2.070
ARIMA(3, 0, 2)	9	22.29	25.72	0.896	1.070	-1.192
ARIMA(3, 0, 1)	10	22.30	25.90	0.965	1.070	-1.232
ARIMA(1, 1, 2)	10	22.82	28.48	0.988	1.095	-1.700
ARIMA(1, 0, 3)	10	23.32	26.43	1.009	1.119	-1.325
ARIMA(0, 1, 0)	10	23.34	31.56	1.010	1.120	-2.315
ARIMA(1, 1, 3)	10	23.45	34.82	1.015	1.126	-3.035

Model	n	MAE	RMSE	NMAE	MASE	R^2_{pred}
ARIMA(0, 1, 3)	10	23.91	30.92	1.035	1.148	-2.182
ARIMA(3, 1, 1)	10	24.23	32.85	1.049	1.163	-2.592
ARIMA(0, 0, 3)	10	24.63	28.60	1.066	1.182	-1.723
ARIMA(2, 1, 3)	9	24.90	35.94	1.000	1.195	-3.279
ARIMA(3, 0, 3)	8	32.05	37.86	1.233	1.538	-3.363
ARIMA(2, 0, 2)	10	46.30	85.44	2.004	2.222	-23.295

3.2 Non-ARIMA references

Model	n	MAE	RMSE	NMAE	MASE	R^2_{pred}
Damped Holt	10	12.56	19.26	0.544	0.603	-0.234
AutoReg(2)	10	13.69	17.53	0.593	0.657	-0.022
AutoReg(1)	10	14.25	18.69	0.617	0.684	-0.162
Additive Holt	10	14.72	21.28	0.637	0.706	-0.507
AutoReg(3)	10	14.82	17.99	0.641	0.711	-0.077
Simple exponential smoothing	10	15.31	18.41	0.663	0.735	-0.128
Historical mean	10	15.31	18.41	0.663	0.735	-0.128
Naive	10	20.40	26.75	0.883	0.979	-1.381

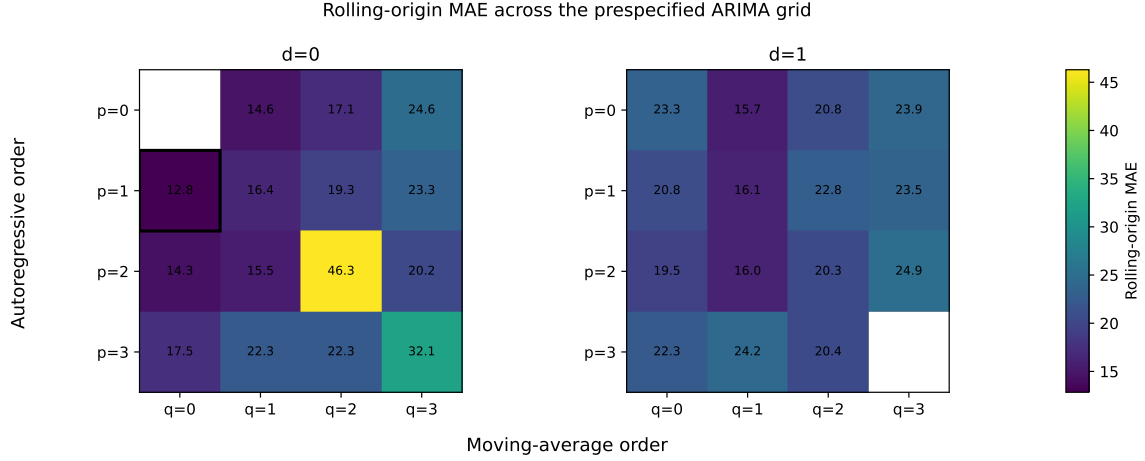


Figure 1: Rolling-origin MAE across the prespecified ARIMA grid. ARIMA(1,0,0) is outlined.

Negative predictive R^2 values are valid: they mean that the squared forecast error exceeded the error of using the validation-window mean. They are not retrospective coefficients of determination.

4 ARIMAX lag sensitivity and coefficients

Specification	MAE	RMSE	R^2_{pred}	MSE skill	p_H	p_P
Heat + prevention ($t - 1$)	10.32	12.93	0.491	0.597	0.061	0.175
Heat + prevention ($t - 2$)	16.83	25.06	-0.911	-0.511	0.001	0.010
Heat + prevention ($t - 3$)	8.79	13.37	0.456	0.570	0.013	0.332
Heat only	9.64	12.85	0.497	0.602	0.099	—
ARIMA(1, 0, 0)	13.97	20.38	-0.265	0.000	—	—

The lag-one model was retained for conditional prediction because it had the lowest RMSE among specifications containing prevention. The lag-two model was retained for coefficient interpretation because it was the structural lag and had the most precise coefficients. No lag is claimed to be causally identified from the national series.

5 Sequential external validation

Model	Year	Observed	Predicted	PI90 low	PI90 high	Std. residual	Upper-tail p
ARIMA(1,0,0)	2024	16	19.22	5.25	64.44	-0.243	0.5961
ARIMA(1,0,0)	2025	63	19.36	5.48	62.97	1.645	0.0499
ARIMAX heat + prevention ($t-1$)	2024	16	23.85	9.40	58.33	-0.717	0.7633
ARIMAX heat + prevention ($t-1$)	2025	63	46.67	19.22	111.38	0.565	0.2861
ARIMAX heat + prevention ($t-2$)	2024	16	21.15	8.88	48.67	-0.539	0.7050
ARIMAX heat + prevention ($t-2$)	2025	63	38.17	16.78	85.29	1.022	0.1533
ARIMAX heat + prevention ($t-3$)	2024	16	21.09	7.97	53.42	-0.478	0.6838
ARIMAX heat + prevention ($t-3$)	2025	63	32.81	13.05	80.38	1.195	0.1161
ARIMAX events + investment ($t-2$)	2024	16	29.98	14.00	62.98	-1.362	0.9133
ARIMAX events + investment ($t-2$)	2025	63	17.44	7.79	37.66	2.765	0.0028

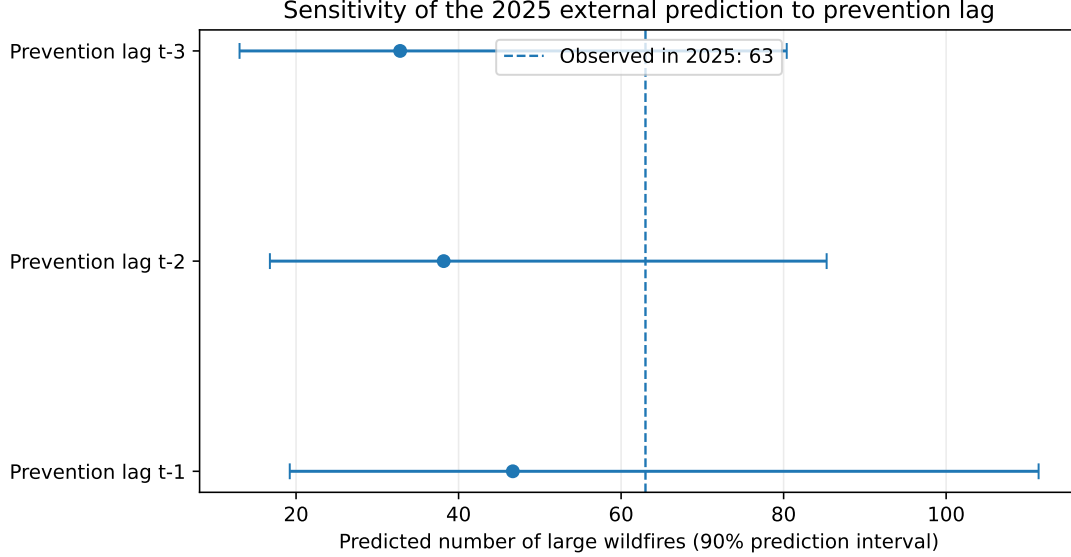


Figure 2: Sensitivity of the 2025 conditional prediction to the prevention lag.

6 Prospective 2026 end-of-August snapshot

The 2026 target was not incorporated into fitting or model selection. ARIMA(1,0,0) was refitted using annual outcomes through 2025 and frozen before comparison with the provisional 2026 season.

Quantity	Point/count	Lower	Upper
Frozen ARIMA(1,0,0) forecast	17.44	4.62	59.54
MITECO provisional, 2 Aug 2026	32	—	—
MITECO/EFFIS estimate, 7 Aug 2026	39	—	—
MITECO balance, 17 Aug 2026	42	—	—
MITECO balance, 31 Aug 2026	44	—	—
Historical completion benchmark from 44	49.9	45.9	68.5

Table 3: Prospective 2026 snapshot. ARIMA limits are the 90% prediction interval. The end-August completion benchmark applies the median and interquartile range of final-to-31-August count ratios from 2015–2024 and is descriptive, not a model forecast.

The prospective sequence progressed from 32 large wildfires on 2 August to an EFFIS-supported estimate of 39 on 7 August, 42 in the mid-August MITECO balance and 44 in the national balance reported after the 31 August follow-up meeting. The corresponding end-August affected forest area was 264,656 ha. At 31 August 2025, the official MITECO report had recorded 60 large wildfires and an EFFIS-supported estimate of 346,443 ha. The 2015–2024 mean count at 31 August was 17.8. Applying the historical median final-to-31-August ratio (1.134) to the current 44 gives a descriptive completion value of 49.9 LWs (interquartile range 45.9–68.5); this is not a model forecast. All 2026 values remain provisional. The predictive lag-one ARIMAX requires 2025 prevention expenditure, which was unavailable in the harmonised series; the value was therefore left missing rather than imputed.

For sensitivity only, the lag-two log-linear equation can be evaluated conditionally because it uses the known 2024 prevention value (7.377 EUR ha⁻¹). The heat coefficient fitted through 2025 was 0.04843 per day ($p = 0.00088$); the prevention coefficient was 0.10752 ($p = 0.0612$), with retrospective $R^2_{\text{fit,log}} = 0.510$.

2026 heatwave-day scenario	Predicted LWs	PI90 low	PI90 high
20	25.61	9.06	69.38
25	32.90	11.59	90.31
30	42.19	14.59	118.69
35	54.03	18.12	157.39
40	69.10	22.25	210.38

Table 4: Conditional 2026 heat-exposure scenarios from the lag-two inferential equation. These are not annual forecasts because the final 2026 heatwave-day total is not yet available.

7 Complete negative-binomial rolling validation

Model	MAE	RMSE	NMAE	NRMSE	MASE	R^2_{pred}	MSE skill	CRPS	Coverage
Events + investment ($t - 2$)	11.48	15.90	0.497	0.688	0.569	0.159	0.373	8.66	0.70
Events + prevention ($t - 2$)	14.40	17.57	0.624	0.761	0.714	-0.028	0.234	10.14	0.60
Endogenous	16.70	20.07	0.723	0.869	0.827	-0.341	-0.000	10.43	0.90
Investment ($t - 2$)	14.97	20.18	0.648	0.874	0.742	-0.356	-0.011	10.87	0.80
Prevention ($t - 2$)	15.62	20.01	0.676	0.866	0.774	-0.333	0.006	11.01	0.80
Heat events	28.05	38.43	1.214	1.664	1.389	-3.916	-2.665	15.11	0.80
Heatwave days + prevention ($t - 2$)	29.32	47.83	1.269	2.071	1.452	-6.613	-4.677	19.16	0.80

8 Bayesian posterior summary

Parameter	Mean	2.5%	97.5%	IRR	$P(< 0)$	$\hat{R}$
Heatwave days (standardised)	0.501	0.173	0.814	1.587	0.003	1.000
Prevention ($t - 2$) (standardised)	0.235	-0.092	0.542	1.095	0.070	1.001
Intercept	2.984	2.661	3.314	—	0.000	1.003
AR(1) coefficient	-0.051	-0.896	0.890	—	0.547	1.007
State SD	0.283	0.016	0.684	—	0.000	1.000
Dispersion	7.943	2.123	20.919	—	0.000	1.001

On the natural scale, ten additional heatwave days had posterior IRR 1.587 (95% credible interval 1.169–2.081). One additional real EUR ha⁻¹ of lag-two prevention had posterior IRR 1.095 (0.966–1.227).

9 PCA loadings

Variable	PC1	PC2
Large-wildfire count	0.574	0.783
Heatwave days	0.886	0.295
Heatwave events	0.911	0.240
Prevention ($t - 2$)	-0.748	0.629
Total investment ($t - 2$)	-0.606	0.758

PC1 and PC2 explained 54.0% and 32.5% of variance, respectively. Regional panel modelling and observational FWI integration are reserved for a separate follow-up study and are not presented as completed analyses in this supplement.